

42904 Cloud Computing and Software as a Service

## Assignment 2: Research Report

# Contract Management System

developed using Force.com platform



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## 1. Introduction

This report presents the development of AKP Contracts, a hypothetical mid-sized Australian financial services organization in charge of Contract Management System Prototype using Force.com Platform as a Service (PaaS). AKP Contracts manages a significant volume of contracts with external vendors, service providers and internal departments, covering areas such as IT services, facilities management, legal advisory and procurement.

After evaluating the organization's existing contract management processes, relying on email correspondence, shared spreadsheets, no employee's tasks tracking and manual document storage, AKP Contracts meets efficiency, accountability or regulatory compliance requirements. It allows us to reduce lost contract processes by delayed email chains; reduced approval processed with minimum visibility and renewal deadlines missed due to absence of automated reminders.

A small prototype for a new Contract Management System has been developed as a cost-effective alternative to existing manual systems, using Force.com PaaS to demonstrate its capabilities in addressing the organization's key business requirements.

| Force.com Login Credentials |                          |
|-----------------------------|--------------------------|
| Username                    |                          |
| Password                    | <input type="password"/> |

## 1.1. Specific Use Cases for the Contract Management System

The required core functionalities of the application include:

- Creating and managing vendor records to centralize all external party information.
- Change status between drafting, pending to submit to tracking contracts records. This will allow Contract Managers to create contracts linked to vendors, record key details such as contract value, start and end dates, status and track each contract through approval lifecycle
- Formal contract review and approval. A window to formally approve the contracts within the same system, with all the processes and dates recorded against each contract.
- Automated email notifications. An email notification when the contract is submitted for review, when an approval decision has been made and when a contract is approaching its renewal date to eliminate manual follow-ups and separate platforms to monitor and track the tasks.
- Task creation and assignment. A task is automatically created and assigned to the Employee when a contract is submitted for review, ensuring a clear and accountable action path without reliance on verbal or email-based reminders.
- Time-based renewal reminders. The system automatically sends a renewal reminder email to the Employee 30 days before a contract's end date, preventing them from expiring unnoticed and being an asset for high-volume management.

Below is a high-level use case diagram illustrating the application's main users. Further specifics will be described in detail throughout this report.

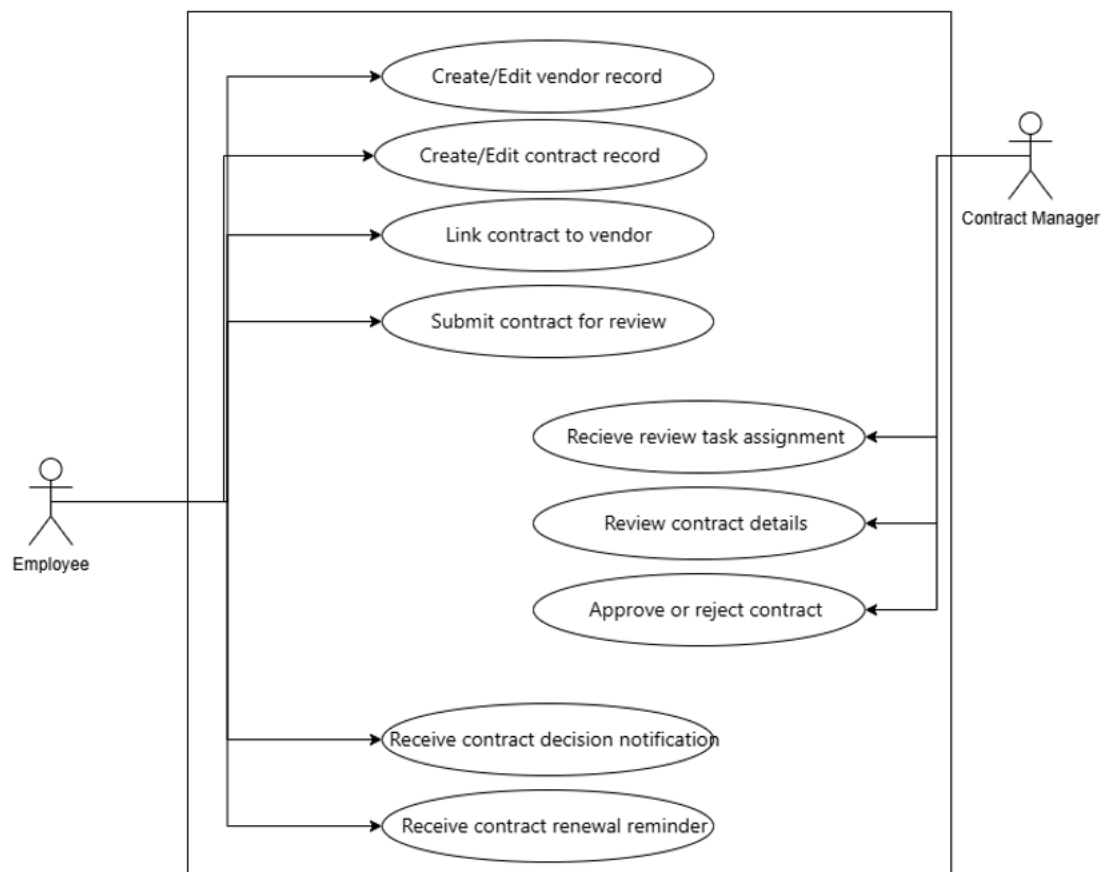
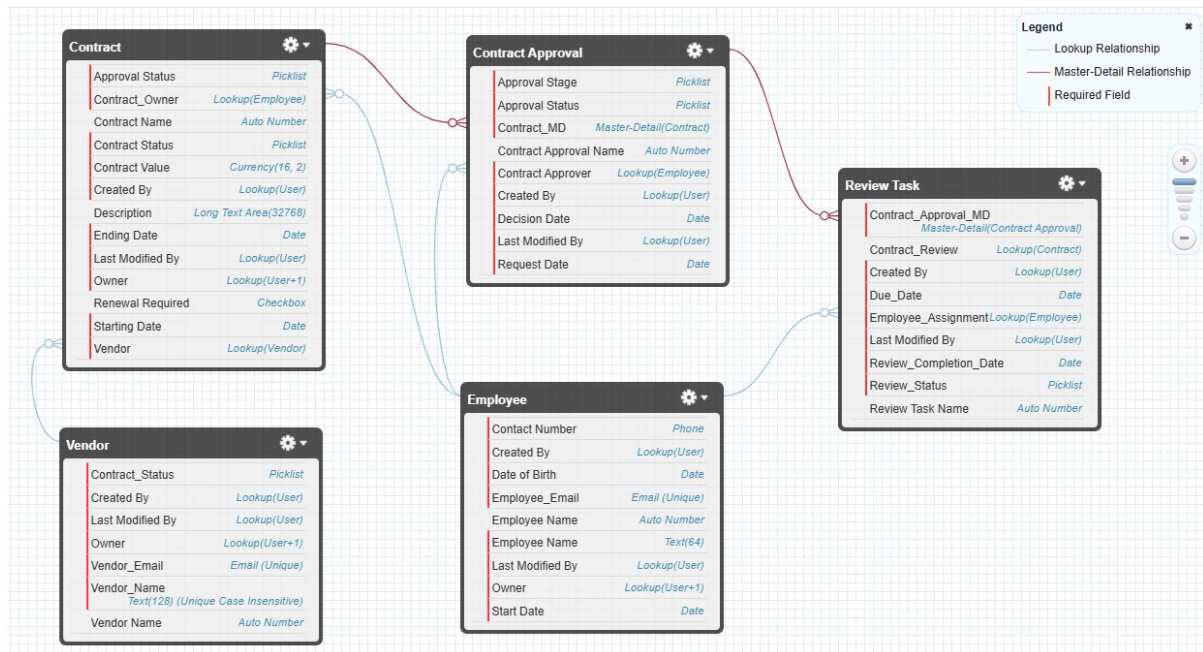


Figure 1. High-level use case diagram for the Contract Management System

## 2. System Modelling

### 2.1. Data Model for the Contract Management System



The application consists of five custom objects used to manage contracts, approvals, vendors, employees, and review activities within the organization.

| Object            | Description                                                                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Employee          | Stores employee details such as employee name, email, contact number, date of birth, and start date. Employees can act as contract owners, approvers, or review task assignees. |
| Vendor            | The Vendor object represents external vendors or service providers linked to contracts. Stores vendor information such as vendor name and vendor email.                         |
| Contract          | Represents the main contract record and stores details such as contract owner, vendor, contract value, contract status, start date, end date, and renewal requirement.          |
| Contract Approval | Represents the approval process of a contract. Stores approval stage, approval status, approver, request date, and decision date.                                               |
| Review Tasks      | Represents review tasks linked to contracts and approvals. Stores assigned employees, due date, completion date, and review status.                                             |

### 2.2. Custom User Profiles for the Contract Management System

## Employee

The Employee profile is intended for regular employees who require limited system access. Users with this profile can view contracts, vendors, and employee information relevant to their role. Employees can also access and update review tasks assigned to them.

- View contract and vendor information.
- Access employee records with limited permissions.
- View and update assigned review tasks.
- Access contract approval information in read-only mode.

|                    | Basic Access                        |                          |                          |                          | Data Administration      |                          |                          |
|--------------------|-------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
|                    | Read                                | Create                   | Edit                     | Delete                   | View All Records         | Modify All Records       | View All Fields          |
| Contracts          | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Contract Approvals | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Employees          | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

|              | Basic Access                        |                          |                                     |                          | Data Administration      |                          |                          |
|--------------|-------------------------------------|--------------------------|-------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
|              | Read                                | Create                   | Edit                                | Delete                   | View All Records         | Modify All Records       | View All Fields          |
| Review Tasks | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Vendors      | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## Contract Manager

The Contract Manager profile is intended for users responsible for managing contracts and approval processes within the organization. This profile provides broader permissions for maintaining contracts, vendors, approvals, and review activities.

- Create and manage contract records.
- Create and manage vendor records.
- Manage contract approval processes.
- Assign and monitor review tasks.
- Update contract statuses and approval stages.

|                    | Basic Access                        |                                     |                                     |                                     | Data Administration      |                          |                          |
|--------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|--------------------------|--------------------------|--------------------------|
|                    | Read                                | Create                              | Edit                                | Delete                              | View All Records         | Modify All Records       | View All Fields          |
| Contracts          | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Contract Approvals | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Employees          | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

|              | Basic Access                        |                                     |                                     |                                     | Data Administration      |                          |                          |
|--------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|--------------------------|--------------------------|--------------------------|
|              | Read                                | Create                              | Edit                                | Delete                              | View All Records         | Modify All Records       | View All Fields          |
| Review Tasks | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Vendors      | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## Organisation-Wide Default (OWD)

|                   |                      |                      |                                     |
|-------------------|----------------------|----------------------|-------------------------------------|
| Contract          | Public Read Only     | Private              | <input checked="" type="checkbox"/> |
| Contract Approval | Controlled by Parent | Controlled by Parent |                                     |
| Employee          | Public Read Only     | Private              | <input checked="" type="checkbox"/> |
| Review Task       | Controlled by Parent | Controlled by Parent |                                     |
| Vendor            | Public Read/Write    | Private              | <input checked="" type="checkbox"/> |

## 2.3. Workflow Diagrams

Three workflow automations were implemented in the Contract Management System to replace manual email-based processes. The full contract even lifecycle is covered, satisfying the requirements email notification, task creation, and time-based workflow automation.

The table below summarizes all three workflow automations and how they map to the assignment requirements:

| Workflow                                    | Assignment Requirement Satisfied                           |
|---------------------------------------------|------------------------------------------------------------|
| Workflow 1 — Contract Submitted for Review  | Email Notification (1 of 2) + Task Creation and Assignment |
| Workflow 2 — Contract Decision Notification | Email Notification (2 of 2)                                |
| Workflow 3 — Contract Renewal Reminder      | Time-Based Workflow                                        |

### Contract Approval Workflow

This workflow is triggered immediately when a Contract record's Status field is changed to 'Pending Review' and the record is saved. The workflow fires two automated actions in sequence:

- Email notification: the system sends an email to the employee informing them that a new contract is awaiting their review.
- Task creation and assignment: the system automatically creates a Review Contract Task and assigns it to the employee, with a due date set to three days from the trigger date, which will need final approval by Contract Manager.

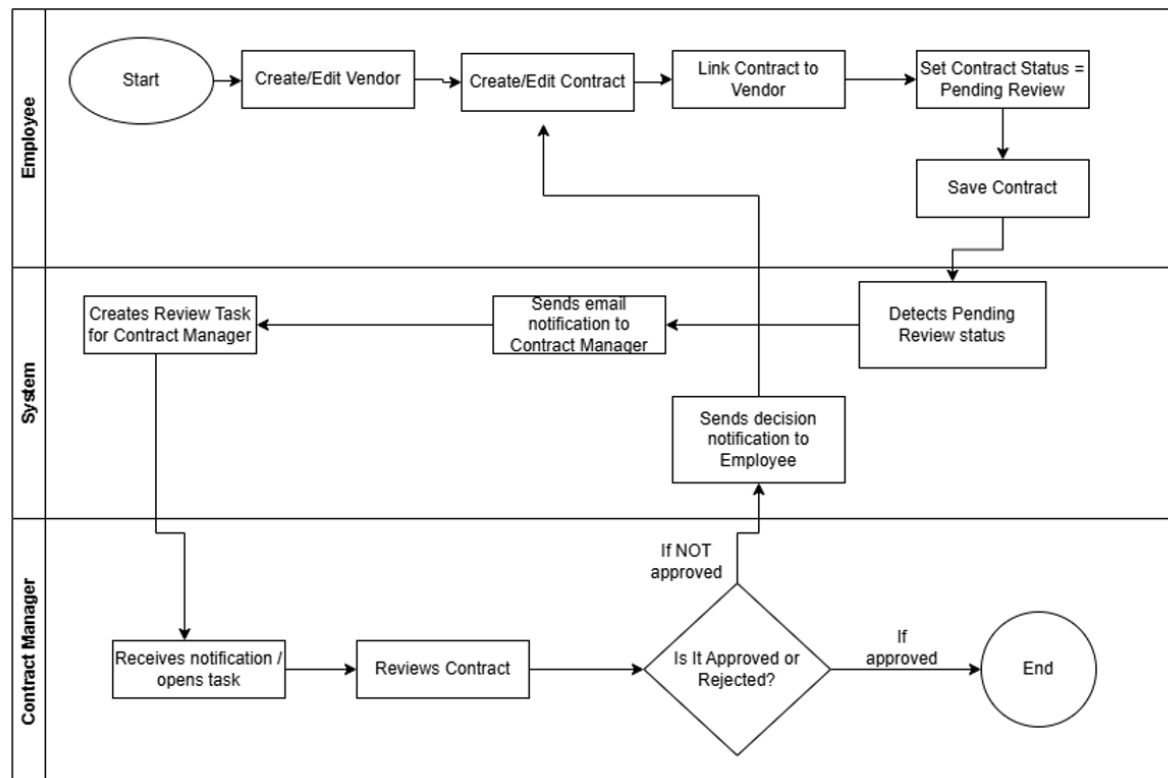


Figure 2. Workflow 1 — Contract Review and Approval Workflow

### Contract Decision Notification Workflow

This workflow is triggered when a Contract Approval record's Decision field is updated to either Approved or Rejected. Once the decision is saved, the system automatically sends an email notification to the Employee who submitted the contract, informing them of the approval outcome.

- If Approved: The Employee receives a contract approval confirmation email, allowing them to proceed with the next stage of the contract process.
- If Rejected: The Employee receives a rejection notification email, including the Contract Manager's comments, allowing them to address the issues and resubmit the contract for review.



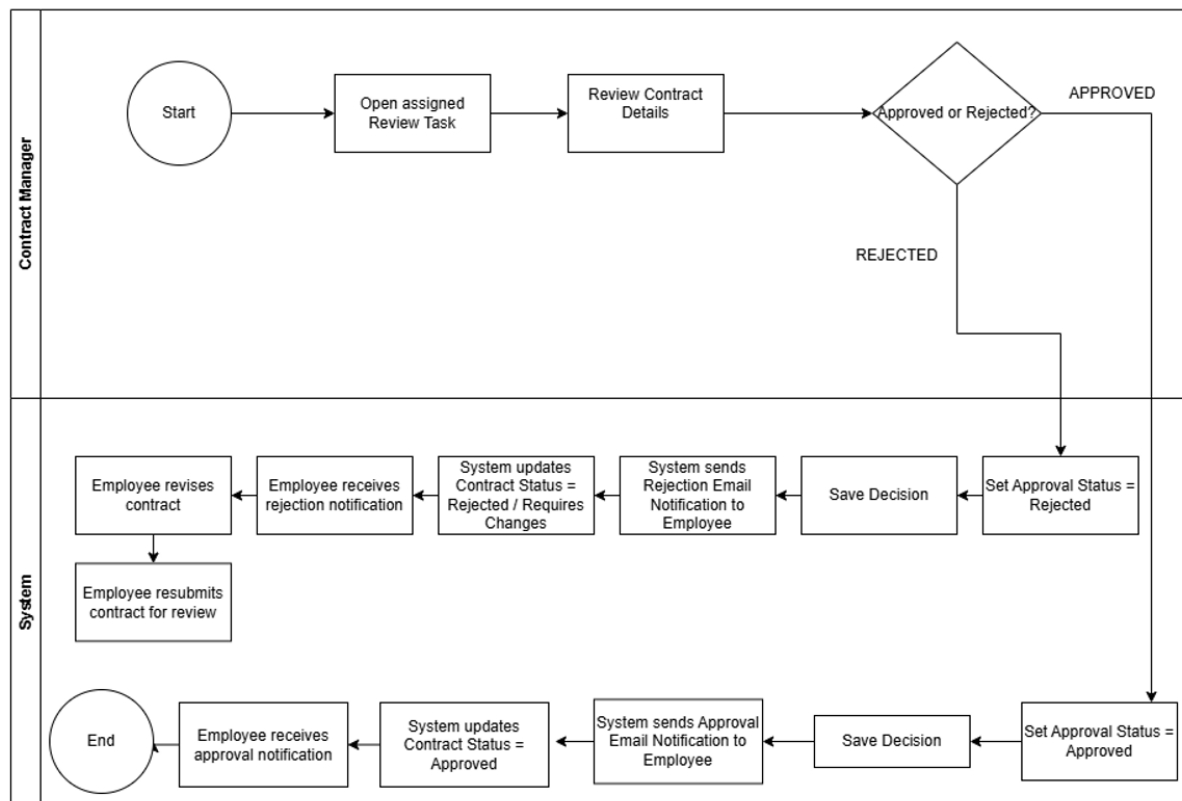


Figure 3. Workflow 2 - Contract Decision Notification

### Contract Renewal Reminder Workflow

This workflow is triggered using a time-based rule connected to the Contract record's Ending Date. When a contract is approaching its expiry date, the system automatically sends a renewal reminder email to the responsible user, such as the Employee.

The reminder prompts the user to review the contract before it expires and decide whether the contract should be renewed, updated, or closed. This workflow satisfies the time-based workflow requirement because the action is triggered by a date field rather than an immediate record update.

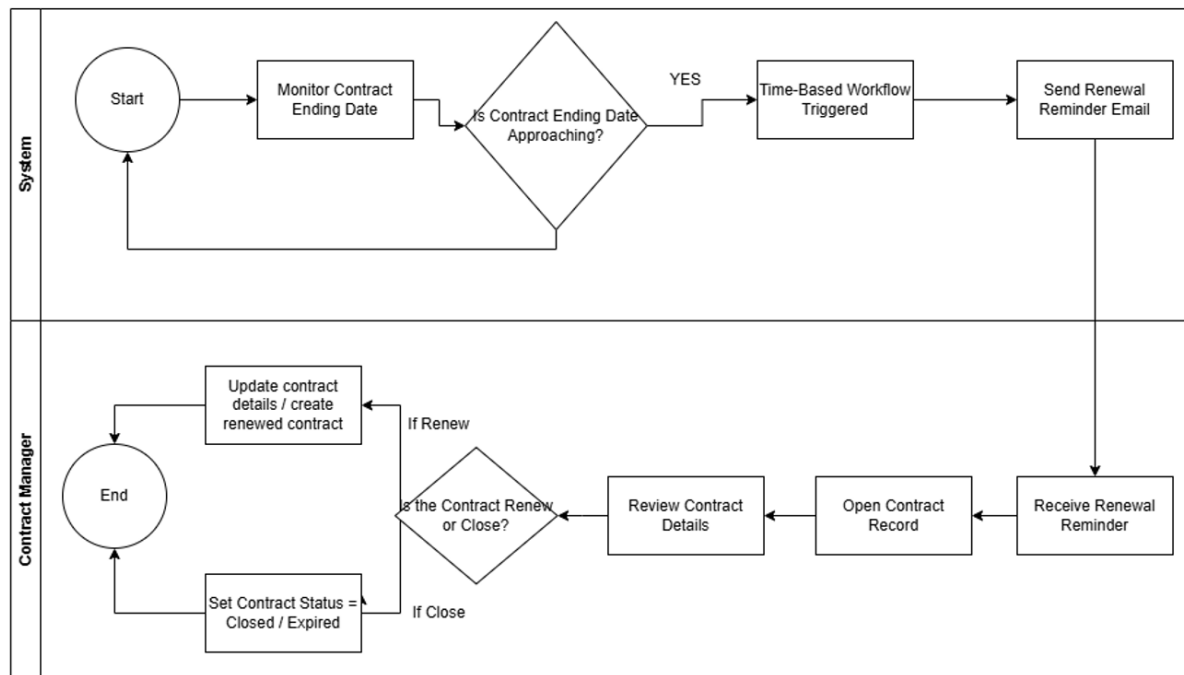


Figure 4. Workflow 3 — Contract Renewal Reminder (Time-Based)

## 2.4. Prototype System Screens and Descriptions

### 2.4.1. Vendor Registration

For contract management system, a complete list of participating vendors are crucial for contract drafting, contract review, and other contract-related activities. As a result, vendors are registered first.

Figure 5. Registering a Vendor

### 2.4.2. Contract Drafting

In practice, to initiate a contract with some vendors requires vendor investigation, contract draft, contract review, meeting with vendor to discuss contract details, contract approval, contract renewal and many more. To get started, employees must create a contract, and specify an active vendor. By leveraging the lookup relationship field type, employees can select the appropriate vendor for the contract.

Figure 6. Creating Contract – Lookup Relationship

The contract object fields are guarded with validation rules. It must have a Contract Value that is greater than 0, an Ending Date later than its starting date, and an active Vendor.

Figure 7. Creating Contract – Validation Rules

### 2.4.3. Submitting a Contract for Review

By selecting “Under Review” in the Contract Status field, the contract will be sent to contract manager for review. The system will monitor the Contract Status field of a contract. Once it turns to “Under Review”, a review task will be assigned to an employee.

Figure 8. Contract Approval Task Assignment

Meanwhile, an Email notification will be sent to the employee informing the task assignment.

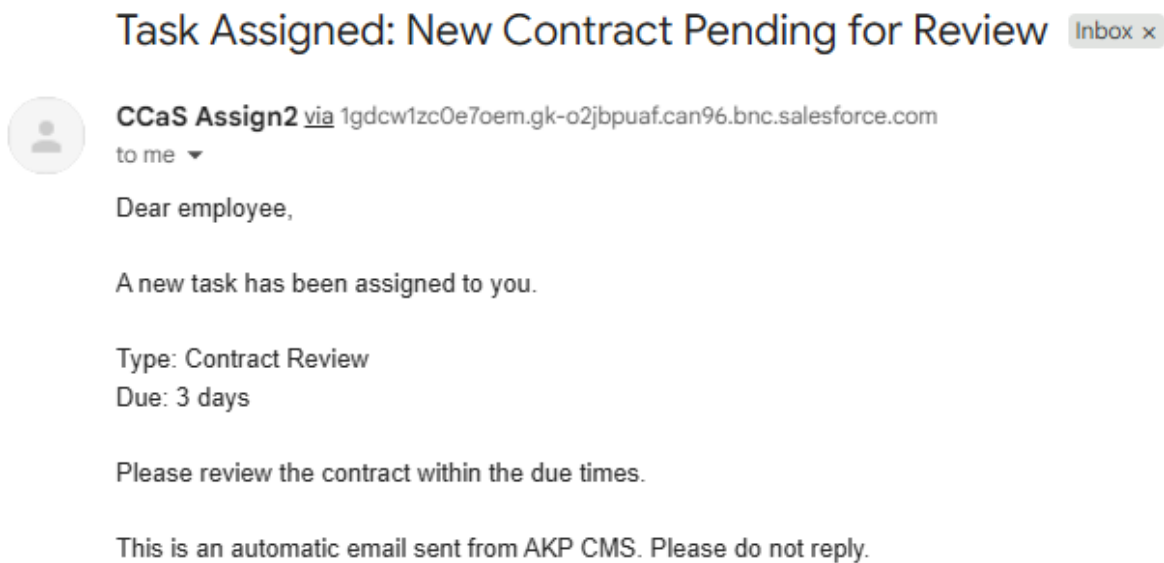


Figure 9. Contract Approval Email Notification

#### 2.4.4. Contract Approval / Rejection

Employee will then review the contract and make a decision on whether the contract should be approved or rejected. Once the decision is made, employee will create a Contract Approval to deliver this decision. Each Contract Approval belongs to the contract being reviewed. Employee will select the contract in the Contract\_MD field which uses a Master-Detail relationship between Contract Approval and Contract.

Contract Approval Edit  
New Contract Approval Help for this Page

**Contract Approval Edit** Save Save & New Cancel

**Contract Approval Details** Required Information

Contract Approver

Request Date

Approval Stage

Approval Status

Decision Date

**Information**

Contract\_MD

Save Save & New Cancel

Figure 10. Contract Approval – Master-Detail Relationship

The system will monitor the Approval Status field. If it is either “Approved” or “Rejected”, the system will send out an email notification to the employee informing the contract review result.

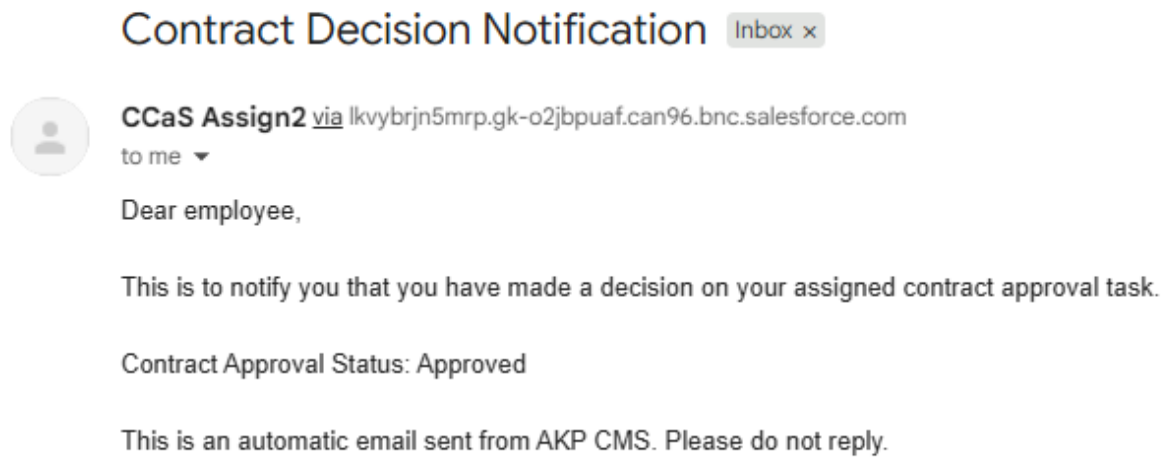


Figure 11. Contract Decision Notification Email

### 2.4.5. Renewal Reminder

When a contract is created, system will monitor its ending date. Depending on the ending date, the system will schedule a renewal reminder email that will be sent to the responsible employee 30 days before the ending date.

#### Time-Based Automations

[Help for this Page](#)

Monitor the queue of pending automations. To filter the list, enter criteria before you click Search. The queue includes scheduled paths for record-triggered flows, schedule-triggered flows, time-dependent workflow actions, scheduled actions for processes built in Process Builder, and flow resume events.

|          |   |          |   |  |     |
|----------|---|----------|---|--|-----|
| --None-- | ▼ | --None-- | ▼ |  | AND |
| --None-- | ▼ | --None-- | ▼ |  | AND |
| --None-- | ▼ | --None-- | ▼ |  | AND |
| --None-- | ▼ | --None-- | ▼ |  | AND |
| --None-- | ▼ | --None-- | ▼ |  | AND |

| <input type="checkbox"/> Record Name | Object   | Workflow Rule, Flow, or Process Name | Scheduled Date     | Created By   | Created Date       | Automation Type  |
|--------------------------------------|----------|--------------------------------------|--------------------|--------------|--------------------|------------------|
| <input type="checkbox"/> C-0003      | Contract | Contract Renewal Reminder Workflow   | 19/5/2026, 5:00 pm | Assign2_CCaS | 17/5/2026, 8:52 pm | Workflow Actions |

Figure 12. Renewal Reminder – Time-based Workflow

## 3. Contributions

All members of the group contributed equally to completing this assignment. The team collaborated throughout the project, meeting regularly to jointly build and test the Contract Management System in Salesforce Force.com, troubleshoot implementation issues, and review each other's work before submission.

| Group Member   | Student ID | Contributions                                                                                                          |
|----------------|------------|------------------------------------------------------------------------------------------------------------------------|
| Adithya Maurya |            | Salesforce: All settings. Report: Section 2.1 Data Model, Section 2.2 Custom User Profiles                             |
|                |            | Salesforce: All settings. Report: Section 2.4 Prototype Screens and Descriptions, Section 4 Conclusion and Future Work |
|                |            | Salesforce: All settings. Report: Section 1 Introduction, Section 2.3 Workflow Diagrams                                |

## 4. Conclusion and Future Work

Developing the AKP Contract Management System using Force.com as a Platform as a Service (PaaS) was a valuable learning experience for the group. At the beginning of the project, the team found the Force.com environment and its large number of configuration options somewhat difficult to navigate. However, after working through the platform, studying the available learning materials and gaining hands-on experience, the team became more confident in using Force.com to design and manage business-oriented applications.

Working within the limitations of our current experience also challenged the group to think carefully about how to implement practical contract management functions using the tools available within the platform. As development progressed, it became clear that Force.com offers significant flexibility for building workflow-driven enterprise systems, particularly through features such as custom objects, workflow automation, validation rules and role-based security settings. Throughout the project, the team frequently discovered additional platform tools and features that could further improve the application if more development time were available.

The current prototype successfully demonstrates core contract management processes such as vendor registration, contract creation, approval workflows, task assignment and automated renewal reminders. However, there are several opportunities for future improvement. Additional features such as reporting dashboards, contract analytics, approval escalation processes, digital signature integration and more advanced validation rules could further enhance the system's functionality and usability.

By adopting the Force.com PaaS platform, AKP Contracts could benefit from improved workflow automation, better contract visibility and reduced reliance on manual email-based processes. Beyond contract management, the organisation could also apply similar PaaS-based solutions in other operational areas such as compliance management, vendor performance tracking, procurement workflows and risk management systems.